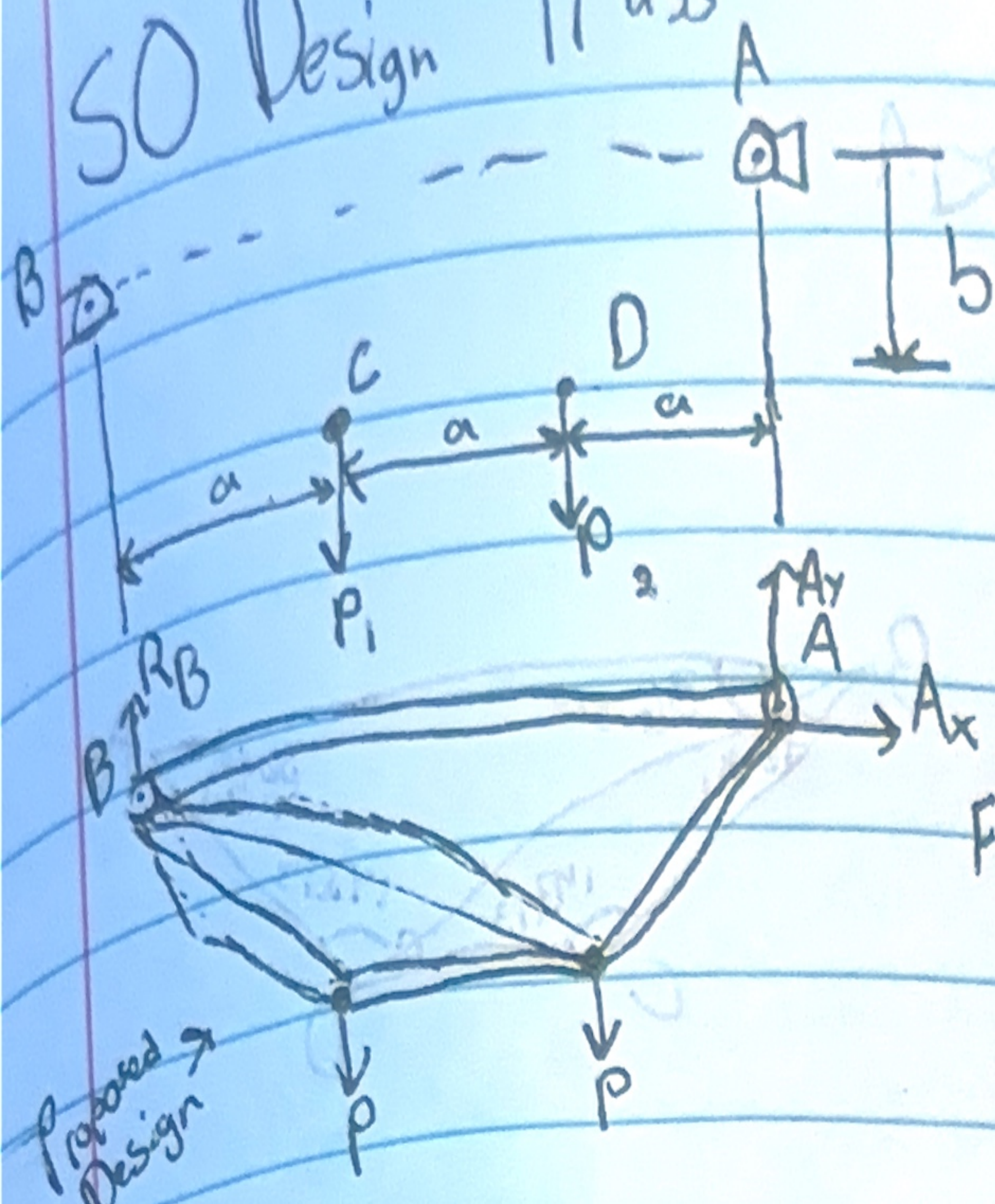
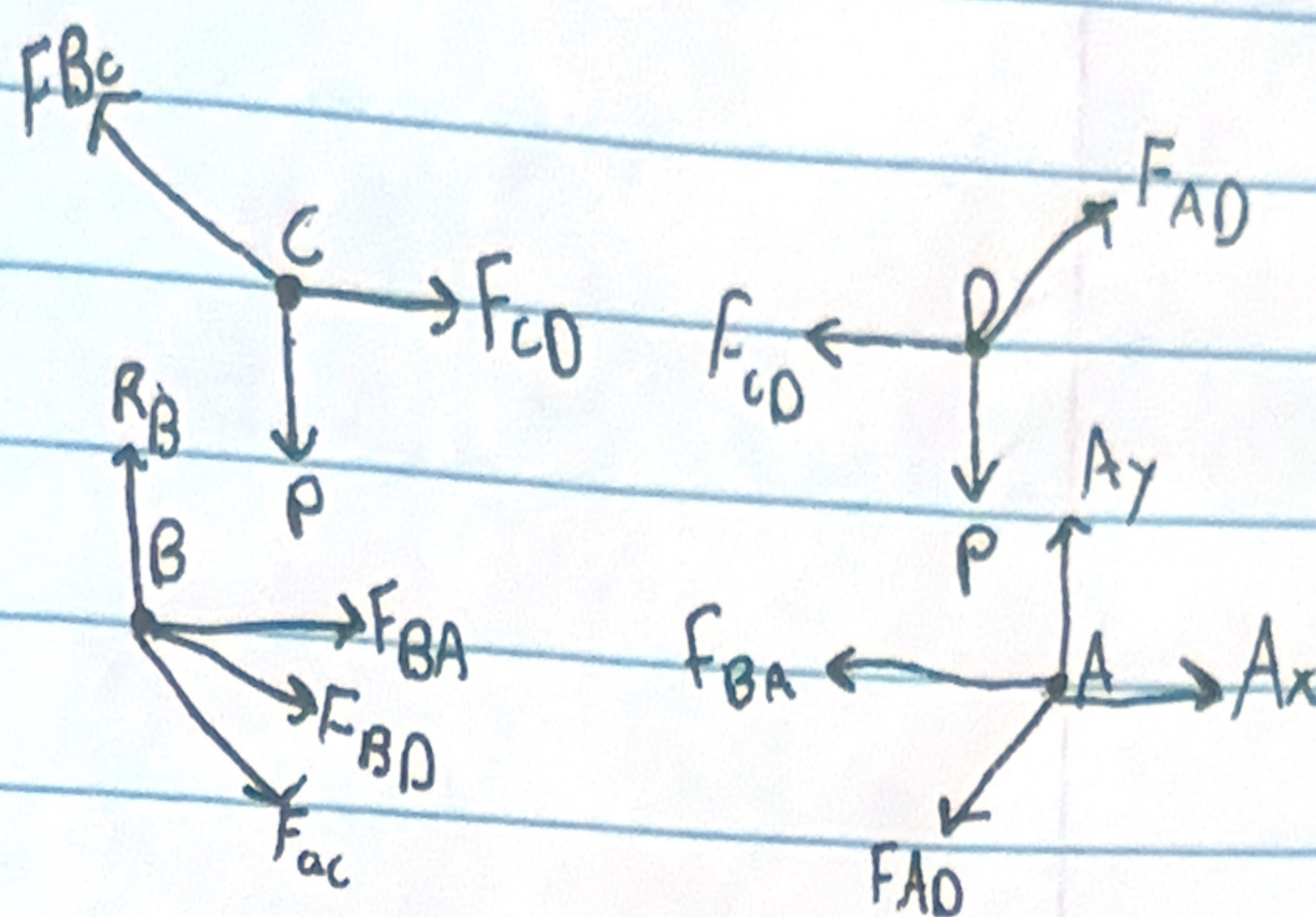


SO Design Truss



$a = 0.4m$ $b = 0.3m$ $P = 25kN$
Length of Members:

$$\begin{aligned} BA &= 3a = 1.2m & BC &= \sqrt{0.4^2 + 0.3^2} = 0.5m \\ CD &= a = 0.4m & AD &= \sqrt{0.4^2 + 0.3^2} = 0.5m \\ BD &= \sqrt{0.8^2 + 0.3^2} = 0.854m \end{aligned}$$



$$R_B = A_y = 25 \quad A_x = 0$$

$$F_{BC} = P\sqrt{a^2 + b^2}/b$$

$$F_{CD} = F_{BC} \left(\frac{a}{\sqrt{a^2 + b^2}} \right) = \frac{Pa}{b}$$

$$F_{BA} = -\frac{Pa}{b}$$

$$F_{BD} = 0$$

$$R_B = A_y = 25kN \quad A_x = 0$$

$$F_{AD} = F_{BC} = 91.67kN \quad F_{BA} = 33.33kN$$

$$F_{CD} = 33.33kN \quad F_{BD} = 0kN$$

Knowns:

Max internal member force = 91.67kN

Safety Factor = 3.5

Yield Strength = 290 MPa

Steel Density = 7850 kg/m³

I chose this design to stay simple. As long as everything is supported and can hold that's what I'm worried about.

It's a non-traditional but effective style of truss.

Unknowns:

Required cross-sectional area

SO Design Truss

$$\sigma_{allow} = F_y / SF$$

$$\sigma_{allow} = 290 \text{ MPa} / 3.5 = 82.9 \text{ MPa}$$

$$A_{req} = 41.67 \text{ kN} / 82.9 = 501.6 \text{ mm}^2$$

$$L_{total} = 4a + 2\sqrt{a^2 + b^2} + \sqrt{a^2 + b^2}$$

$$\text{Weight of truss} = \rho \cdot A_{req} \cdot L_{total}$$

KNOWN:

Largest force through a single pin

$$SF_{pin} = 4$$

$$\tau_v = 170 \text{ ksi}$$

$$\rho_{pin} = .28 \text{ lb/in}^3$$

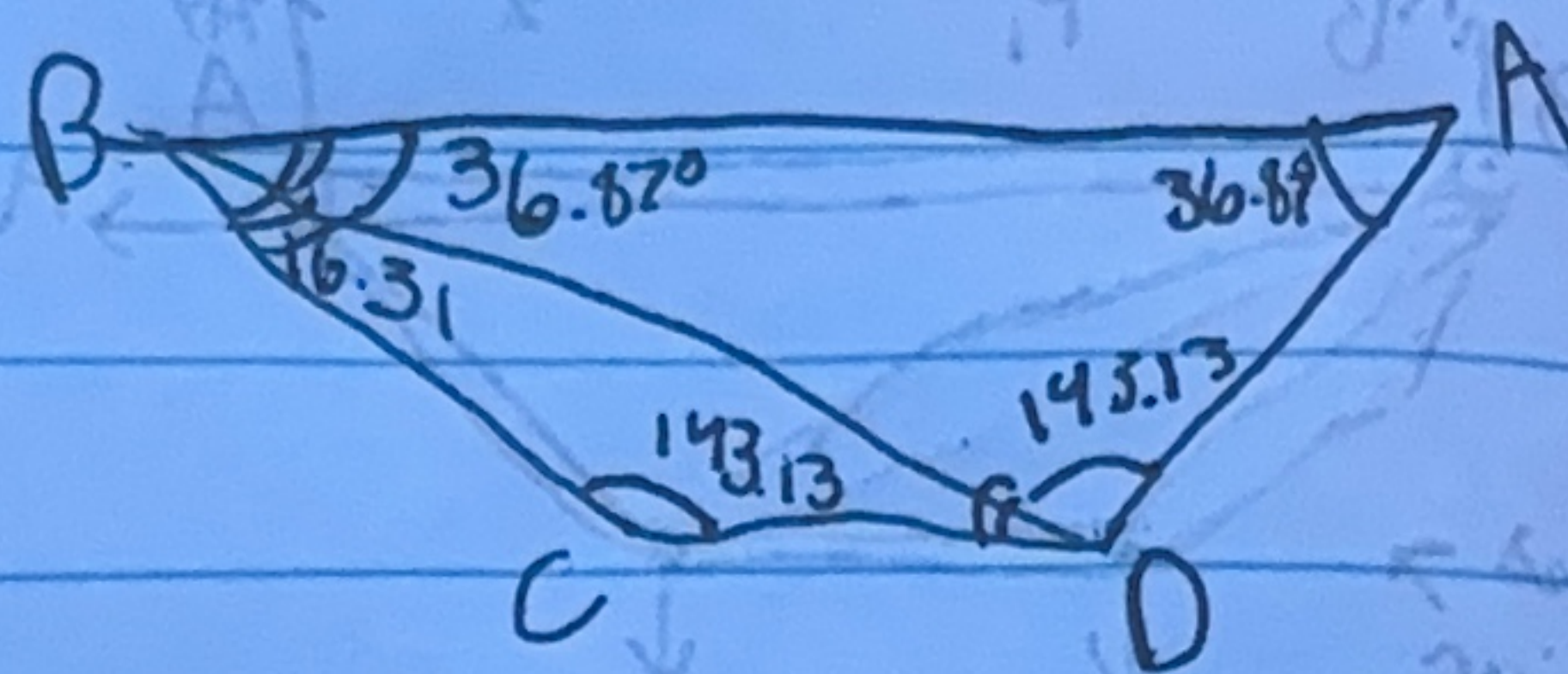
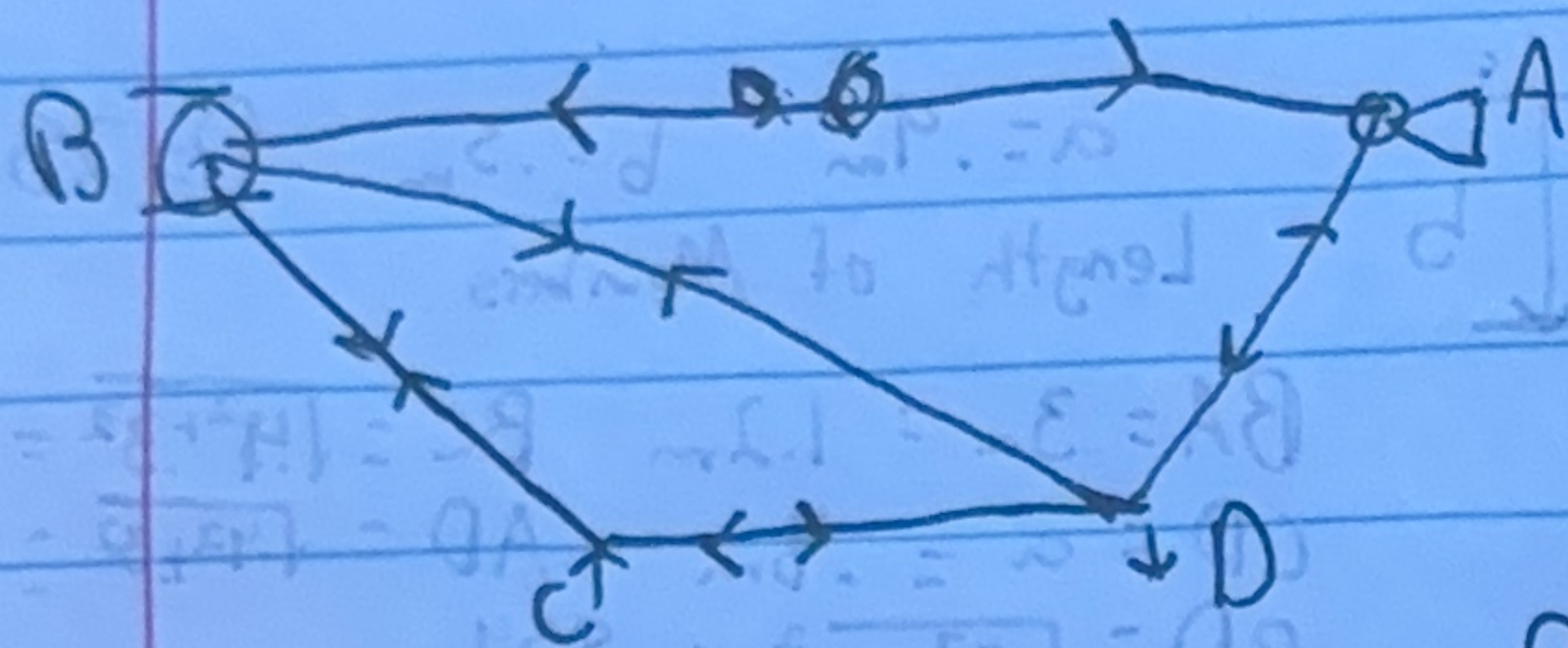
Each pin is

checked for single

Shear against the largest force any pin must transfer.

The pin is modeled

as a short cylindrical shaft loaded by the force across one shear plane.



$$0 = \sum F_x = A_x = 0$$

$$F_{BC} = 41.67 \text{ kN}$$

$$F_{CD} = 41.67 \text{ kN}$$

$$F_{DA} = 41.67 \text{ kN}$$

$$0 = \sum F_x = A_x = 0$$

$$F_{BC} = 41.67 \text{ kN}$$

$$F_{CD} = 41.67 \text{ kN}$$

$$F_{DA} = 41.67 \text{ kN}$$

$$F_{DB} = 41.67 \text{ kN}$$

$$\text{Max internal member force} = 41.67 \text{ kN}$$

$$\text{Factor} = 3.5$$

$$\text{Strength} = 147.1 \text{ kN}$$

$$\text{Density} = 7850 \text{ kg/m}^3$$

Known:

Required cross-sectional area

SO Design Truss

$$\sigma_{allow} = F_y / SF$$

$$\sigma_{allow} = 290 \text{ MPa} / 3.5 = 82.9 \text{ MPa}$$

$$A_{req} = F_{max} / \sigma_{allow} = [R \sqrt{a^2 + b^2} / B] \cdot SF / F_y$$

$$A_{req} = 41.67 \text{ kN} / 82.9 = 502.8 \text{ mm}^2$$

$$A_{req} = \pi d^2 / 4 = 25.3 \text{ mm}^2 \text{ or } 1 \text{ in diameter}$$

$$L_{total} = l_a + 2 \sqrt{a^2 + b^2} + \sqrt{a^2 + b^2} = 3.454 \text{ m}$$

$$\text{Weight of truss} = \rho \cdot A_{req} \cdot L_{total} = 7850 \text{ kg/m}^3 \times 502.8 \times 10^{-6} \text{ m}^2 \times 3.454 \text{ m} = 13.6 \text{ kg} \approx 30 \text{ lb}$$

KNOWNs:

Largest force through a single pin = 41.67 kN

SF pin = 4

$\tau_r = 170 \text{ ksi}$

$\rho_{pin} = .28 \text{ lb/in}^3$

Un Knowns:

Pin Cross-Sectional area

diameter and pin weight

Each pin is

checked for single

shear against the largest force any pin must transfer.

The pin is modeled

as a short cylindrical shaft loaded by the force across one shear plane.

$F_{BC} = 41.67 \text{ kN}$

C Pin

$F_{CD} = 33.33 \text{ kN}$

$P = 25 \text{ kN}$

$R_B = 25 \text{ kN}$

B Pin

$F_{AB} = 33.33 \text{ kN}$

$F_{BO} = 0$

$F_{BC} = 41.67 \text{ kN}$

F_{max}

Pin

F_{max}

$F_{CD} = 33.33 \text{ kN}$

$P = 25 \text{ kN}$

$F_{AD} = 41.67 \text{ kN}$

$A_y = 25 \text{ kN}$

$F_{BA} = 33.33 \text{ kN}$

$A_x = 0$

$F_{AD} = 41.67 \text{ kN}$

$$\tau_{allow} = \tau_y / SF_{pin} \quad A_{pin} = F_{max} / \tau_{allow} = [P\sqrt{a^2 + b^2} / b] \cdot \frac{SF_{pin}}{\tau_y}$$

$$\tau_{allow} = 170 \text{ ksi} / 4 = 42.5 \text{ ksi} (293 \text{ MPa})$$

$$A_{pin} = 41.67 \text{ kN} / 293 \text{ MPa} = 142.2 \text{ mm}^2$$

$$d_{pin} = \sqrt{(4 \cdot A_{pin} / \pi)} \approx 13.5 \text{ mm} \approx .53 \text{ in}$$

$$\text{Weight of pins} = 4 \times \rho_{pin} \times A_{pin} \times L_{pin} = 4 \times .281 \text{ lb/in}^3 \times .220 \text{ in}^2 \times 2 \text{ in} \\ 0.491 \text{ lb} (= .22 \text{ kg})$$

This week I made a truss in Cad. I found it a little difficult to design the truss in Cad as I figured it up on paper a lot faster. I found the actual weight has little effect on the load on the pins.